Mini-Project Report

When Machine Learning Fails:

Diagnosing and Repairing Class Imbalance Failure

on the Online Shoppers Purchasing Intention Dataset

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1. Research Question and Chosen Dataset

1.1 Dataset Overview

The dataset used in this project is the **UCI Online Shoppers Purchasing Intention Dataset** (UCI Repository ID: 468). It records 12,330 web browsing sessions collected from an e-commerce website, where each session is described by 17 features capturing user navigation behavior, temporal context, and visitor characteristics. The classification target is the binary variable Revenue, which indicates whether the session concluded with a purchase (True, coded as 1) or did not (False, coded as 0).

The dataset contains a mix of numerical and categorical features. Numerical features include page interaction metrics such as Administrative_Duration, Informational_Duration, ProductRelated_Duration, BounceRates, ExitRates, and PageValues, as well as temporal markers such as SpecialDay and session counts. Categorical features include Month, VisitorType, and Weekend. No missing values were found in the dataset after initial inspection, and 125 duplicate rows were identified and removed before any analysis.

Class distribution: The critical property of this dataset with respect to this project is its pronounced class imbalance. Approximately **84.5%** of sessions correspond to non-purchase outcomes (Revenue = False) and only **15.5%** correspond to purchase outcomes (Revenue = True). The minority class — purchase sessions — is precisely the business-relevant class, because it is these sessions that generate revenue and drive the commercial objective of the e-commerce platform.

Table 1 — Dataset Summary

Dataset	Task	Target	Minority Class	Failure Mode
Online Shoppers Purchasing Intention	Binary classification	Revenue	Revenue = True (~15.5%)	Class imbalance

1.2 Why Class Imbalance is a Meaningful Failure Mode for This Dataset

Class imbalance constitutes a particularly insidious failure mode because it is not directly visible in global performance metrics. A classifier that predicts every session as a non-purchase would immediately achieve an accuracy of approximately 84.5%, while detecting exactly zero purchase sessions. This is not a hypothetical edge case: it is a common practical collapse observed with real-world classifiers trained on skewed distributions without corrective measures.

The failure is especially meaningful in an e-commerce context. The cost asymmetry between false negatives and false positives is significant: a false negative (a purchase session incorrectly classified as non-purchase) represents a missed business opportunity, whereas a false positive (a non-purchase session incorrectly classified as a purchase) might result in a misallocated marketing action, which is generally less costly. This asymmetry justifies treating recall for the minority class as the primary diagnostic metric.

The Online Shoppers dataset is well-suited for this investigation because it presents this imbalance naturally, without requiring artificial manipulation, and because the minority class carries clear semantic meaning tied to the core business objective.

1.3 Research Question

Does class imbalance in the Online Shoppers Purchasing Intention dataset cause non-linear classifiers to achieve acceptable global performance while failing to correctly detect the minority purchase class Revenue=True, and can imbalance-aware corrections improve minority-class recall without destroying precision?

This question is structured to admit a falsifiable answer. The first clause — concerning the failure — is answered by the controlled experiment described in Section 3. The second clause — concerning the correction — is answered by the before-and-after evaluation described in Section 4.

ML Models: In accordance with the course requirements, no linear model is used anywhere in this project. All classifiers belong to non-linear families: ensemble tree-based methods (Random Forest, LightGBM, AdaBoost with decision-tree base estimators) and a feed-forward neural network (MLPClassifier).

2. Reference Model and Observed Symptom

2.1 Experimental Setup — Notebook 1

The reference failure is established in Notebook 1. Three non-linear classifiers are trained on the dataset without any imbalance correction: **Random Forest**, **Multi-Layer Perceptron (MLP)**, and **LightGBM**. Critically, no SMOTE resampling is applied, no class weighting is introduced, and no decision threshold tuning is performed. These restrictions are intentional: the goal is to expose the failure in its natural form.

2.1.1 Preprocessing Protocol

The following preprocessing steps were applied in strict compliance with the anti-leakage protocol described in Section 2.2:

- **log1p transformation:** Applied to five heavily right-skewed behavioral variables (Administrative_Duration, Informational_Duration, ProductRelated_Duration, ProductRelated, PageValues) before any split. This is a deterministic, parameter-free transformation that does not learn from the data distribution.
- **StandardScaler:** Fitted exclusively on the training set and applied to all numerical features to normalize scale differences.
- **OneHotEncoder (handle_unknown='ignore'):** Fitted exclusively on the training set and applied to categorical features (Month, VisitorType, Weekend).
- **Revenue encoding:** The target variable was converted from boolean to binary integer (0/1) prior to the split.

2.1.2 Anti-Leakage Split Protocol

The dataset was divided into three non-overlapping stratified splits: **60% training (≈7,321 sessions), 20% validation (≈2,440 sessions)**, and **20% test (≈2,440 sessions)**. Stratification by Revenue was applied to preserve the imbalance ratio in each split, consistent with the failure mode under study. All learned transformations (StandardScaler, OneHotEncoder) were fitted strictly on the training set, and validation and test sets were transformed using parameters derived exclusively from training data. The test set was kept completely untouched until the final evaluation step.

2.1.3 Model Selection Strategy

Hyperparameter configurations for each model were evaluated on the **validation set only**. Model selection was performed by maximizing **PR-AUC (Average Precision Score)** on the validation set, because this metric is more informative than ROC-AUC in the presence of class imbalance. The selected model was then evaluated exactly once on the test set.

2.2 Observed Symptom

The class imbalance failure appears clearly in the validation results of the three untouched non-linear models. All models reach a high global accuracy, around **89.6% to 90.3%**, but their recall on the minority class Revenue=True remains much lower. For example, **Random Forest** obtains an accuracy of **0.9033**, but its recall for actual buyers is only **0.5302**. This means that although the model looks strong globally, it detects only about **53% of purchase sessions**. The same pattern appears for **MLP**, with **0.8955 accuracy** and **0.5617 recall**, and for **LightGBM**, with **0.9009 accuracy** and **0.6011 recall**.

This shows that the models do not fail randomly. They fail asymmetrically: they are much better at recognizing the majority class Revenue=False than the minority class Revenue=True. The best model by PR-AUC on validation is **LightGBM**, with **PR-AUC = 0.7388**, but even this model reaches only **0.6011 recall** for buyers. Similarly, Random Forest has a strong **ROC-AUC = 0.9213** and **PR-AUC = 0.7385**, but its minority recall remains limited at **0.5302**. These results show that global ranking metrics and accuracy can hide a weaker ability to identify the business-relevant class.

The central symptom is therefore an **accuracy masking effect**. Since most sessions correspond to non-purchases, a model can achieve approximately **90% accuracy** while still missing a large fraction of real buyers. In this context, the important failure is not low overall performance, but the gap between global accuracy and minority-class detection. This is why the diagnosis must focus on **Recall_true**, **F1_true**, and **PR-AUC**, rather than accuracy alone.

Table 2 — Comparison of Untouched Non-Linear Models on the Validation Set

Model	Accuracy	Precision (True)	Recall (True)	F1 (True)	ROC-AUC	PR-AUC	Time (s)	Size (KB)
Random Forest	0.9033	0.7799	0.5302	0.6313	0.9213	0.7385	1.54	2727.3
MLP	0.8955	0.7086	0.5617	0.6266	0.9257	0.7154	9.24	1153.9
LightGBM	0.9009	0.7179	0.6011	0.6543	0.9268	0.7388	0.30	242.5

Confusion matrix interpretation: The confusion matrix of the best untreated model, **LightGBM**, clearly reveals the asymmetric failure pattern. On the test set, the model reaches a high global accuracy of **0.9033**, but it still misses **153 out of 382 actual buyers**, which corresponds to **40.1% of the Revenue=True class**. In other words, almost **4 out of 10 purchase sessions** are incorrectly classified as non-purchase sessions. This is the main symptom of the class imbalance failure: the model performs very well on the majority class Revenue=False, with a recall of **0.96** for non-purchases, but much less effectively on the minority class Revenue=True, where recall drops to **0.5995**. In business terms, each false negative represents a real purchase session that the system failed to detect, meaning a potential buyer would not receive appropriate targeting, engagement, or follow-up.

Precision-Recall curve interpretation: The Precision-Recall curve confirms the same failure from another angle. The best untreated model obtains a **PR-AUC of 0.7482** on the test set, which is clearly better than the random baseline of approximately **0.155**, corresponding to the positive class rate in the dataset. However, this value remains far from the ideal value of **1.0**, showing that the model still struggles to maintain high precision while increasing recall for Revenue=True. At the default decision threshold, the model reaches a precision of **0.7340** and a recall of only **0.5995** for buyers. This means that detecting more buyers would likely require lowering the threshold, but doing so would increase false positives and reduce precision. The PR curve therefore shows the central trade-off of this imbalanced task: the model has learned useful ranking information, but its decision boundary remains biased toward the majority class, making minority-class detection incomplete.

Key conclusion of Section 2: The untreated non-linear models exhibit the expected class imbalance failure pattern. The model with highest accuracy does not deliver the most business value. The correct diagnostic question is not what is the accuracy? but rather how many purchase sessions are being missed?

3. Causal Hypothesis and Controlled Experiment

3.1 Causal Hypothesis

A weak recall for the minority class could, in principle, be caused by multiple factors: insufficient model capacity, poor feature representation, noisy labels, or class imbalance. The following causal hypothesis attributes the symptom to a specific cause:

The low recall for Revenue=True is caused by the underrepresentation of purchase sessions in the training data. Because the model observes many more non-purchase examples than purchase examples during training, the learned decision boundary becomes biased toward predicting the majority class Revenue=False. This bias persists even in powerful non-linear models because no correction is applied to counteract the distributional asymmetry.

3.2 Falsifiability Criterion

The hypothesis is falsifiable because it predicts a measurable degradation when the imbalance is artificially intensified. If the low recall on Revenue=True is mainly caused by the scarcity of positive examples during training, then progressively reducing the number of positive training examples while keeping the validation set fixed should generally reduce the model’s ability to detect buyers.

This prediction is supported by the controlled degradation results. When **100%** of the positive training examples are retained, the model obtains a Recall_true of **0.575** and an F1_true of **0.613**. When only **75%** of the positive examples are kept, recall drops to **0.499**. At **50%**, recall decreases further to **0.451**. The degradation becomes much more severe when the positive class becomes very rare: at **25%** retained positives, recall falls to **0.186**, and at **10%**, it collapses to only **0.066**. In other words, when the training positive rate decreases from **15.64%** to **1.81%**, the model almost stops detecting purchase sessions.

The same trend is visible for F1_true, which decreases from **0.613** at 100% positives to **0.582**, **0.567**, **0.304**, and finally **0.121** at 10%. This confirms that the model is not only losing recall, but also losing balanced minority-class performance. PR_AUC remains relatively stable, between **0.641** and **0.669**, which suggests that the model still preserves some ranking ability. However, the decision at the default threshold becomes increasingly biased against Revenue=True, as shown by the strong collapse in recall and F1-score.

Therefore, the falsifiability criterion is satisfied: as positive examples are progressively removed, the model's ability to detect the minority class degrades strongly. If recall and F1-score had remained stable despite the removal of positive examples, the imbalance hypothesis would have been weakened. Instead, the results support the conclusion that class imbalance is a major contributor to the observed minority-class detection failure.

3.3 Controlled Experiment Design

The controlled experiment is implemented in **Notebook 2: SMOTE + AdaBoost — Imbalance Correction**. Its design follows the logic of a **dose-response experiment**: the independent variable is the proportion of positive training examples retained, while the dependent variables are the minority-class detection metrics measured on a fixed validation set.
The experiment was designed to isolate the effect of class imbalance. The validation set remained fixed across all runs, and the test set was not used. The model family was also kept constant: every run used an AdaBoost classifier with a DecisionTreeClassifier base estimator. The preprocessing pipeline, feature set, negative training examples, and random seed were held fixed. The only variable that changed was the number of Revenue=True examples available during training.

Five degradation conditions were tested: **100%**, **75%**, **50%**, **25%**, and **10%** of the positive training examples retained. At the control condition, **100%**, the training set contained **1145 positive examples** and **6178 negative examples**, corresponding to a positive training rate of **15.64%**. As the positive class was progressively reduced, the positive training rate dropped to **12.19%**, **8.47%**, **4.42%**, and finally **1.81%**.

No correction was applied during this controlled experiment. The model was trained **without SMOTE**, **without class weighting**, and **without threshold tuning**. This is essential because the goal of the experiment was not to test a repair method, but to measure the effect of imbalance itself. Applying a correction would have mixed the cause and the remedy in the same experiment.
The results show a clear dose-response pattern. As the positive class becomes rarer, Recall_true decreases from **0.575** to **0.066**, and F1_true decreases from **0.613** to **0.121**. The strongest collapse occurs between **50%** and **10%** of positive examples retained, where the model becomes increasingly unable to identify buyers. This confirms that the scarcity of minority-class examples directly harms the model's ability to detect Revenue=True.

This experiment is controlled because only one variable changes across conditions: the availability of positive training examples. Since the validation set, model family, preprocessing protocol, feature set, negative examples, and seed remain unchanged, the observed degradation in minority-class recall can be attributed to the increasing class imbalance rather than to unrelated confounding factors.

Why this is a proper control

This experiment is a proper controlled test because only one variable changes across the five conditions: the number of positive training examples, corresponding to the minority class Revenue=True. All negative examples are kept, and the validation set remains fixed. The model family, feature set, preprocessing pipeline, evaluation metrics, and random seed are also kept identical across all runs.
This design isolates the independent variable: the availability of positive examples during training. Therefore, if Recall_true and F1_true decrease when fewer positive examples are retained, the degradation can be attributed to the increasing class imbalance rather than to a change in model architecture, preprocessing, evaluation set, or random protocol.
The results show exactly this pattern. When all positive examples are kept, the model reaches a recall of **0.575** on Revenue=True. As the positive class becomes rarer, recall progressively decreases to **0.499**, **0.451**, **0.186**, and finally **0.066** when only 10% of positive examples are retained. The same trend appears for the F1-score, which drops from **0.613** to **0.121**. This confirms that reducing the availability of minority-class examples directly weakens the model's ability to detect purchase sessions.

Table 3 — Controlled Degradation Experiment Results

Positives Kept (%)	Actual Positive Rate in Training (%)	Recall (Revenue=True)	F1 (Revenue=True)	PR-AUC
100%	15.64%	0.575	0.613	0.646
75%	12.19%	0.499	0.582	0.641
50%	8.47%	0.451	0.567	0.669
25%	4.42%	0.186	0.304	0.656
10%	1.81%	0.066	0.121	0.653

3.4 Interpretation of the Experiment

The controlled degradation experiment provides strong support for the causal hypothesis. As the proportion of positive training examples decreases, the model becomes progressively worse at detecting the minority class Revenue=True. In the control condition, where **100%** of the positive examples are retained, the training set contains **15.64%** positives and the model reaches a Recall_true of **0.575**. When only **75%** of the positives are kept, the positive rate drops to **12.19%** and recall decreases to **0.499**. At **50%**, the positive rate becomes **8.47%**, and recall decreases again to **0.451**.
The degradation becomes much more severe when the minority class becomes extremely rare. With only **25%** of positive examples retained, the positive rate falls to **4.42%**, and Recall_true collapses to **0.186**. At the most extreme condition, where only **10%** of positive examples are kept, the positive rate is only **1.81%**, and recall drops to **0.066**. This means that the model detects only about **6.6%** of actual purchase sessions under the strongest imbalance condition.
The same trend is observed for F1_true. It decreases from **0.613** at 100% positives to **0.582**, **0.567**, **0.304**, and finally **0.121** at 10% positives. This confirms that the degradation is not limited to recall: the overall quality of minority-class detection also deteriorates sharply. In contrast, PR_AUC remains relatively stable, between **0.641** and **0.669**. This suggests that the model still preserves some ranking ability, but its default classification behavior becomes increasingly biased toward the majority class as positive examples become scarce.
This evidence is stronger than a simple baseline observation. Merely observing that the original dataset contains only about **15.5%** positives and that the baseline model has limited recall would

only show a correlation between imbalance and poor minority-class detection. The degradation experiment goes further: it creates a controlled dose-response relationship. As the hypothesized cause intensifies — fewer positive training examples — the hypothesized effect worsens — lower recall and F1-score for Revenue=True.
The curve is not perfectly monotonic for every metric, especially PR_AUC, which slightly increases at the 50% condition. This does not invalidate the hypothesis, because model training and random subsampling can introduce small fluctuations. What matters is the global trend: Recall_true falls from **0.575** to **0.066**, and F1_true falls from **0.613** to **0.121**. A completely flat recall curve would have challenged the hypothesis, but the observed collapse strongly supports it.
The **100% condition** serves as the control condition, representing the original training imbalance without artificial degradation. The other conditions are treatments where the scarcity of positive examples is intensified. The contrast between **100% positives** and **10% positives** provides the strongest evidence: when the positive training rate decreases from **15.64%** to **1.81%**, recall collapses by more than **50 percentage points**. This supports the conclusion that class imbalance is not only associated with low recall, but is causally involved in the minority-class detection failure under this controlled experimental setup.

4. Proposed Correction and Evaluation

4.1 First Correction: SMOTE + AdaBoost + classification

4.1.1 Why this correction targets the cause

The causal mechanism identified in the controlled experiment is the underrepresentation of the positive class Revenue=True during training. When fewer positive examples are available, the model becomes increasingly biased toward the majority class Revenue=False, and the recall on purchase sessions collapses. Therefore, the correction must act before or during training, not only after training. A simple threshold adjustment could increase recall, but it would not change the fact that the model was trained on a distribution dominated by non-purchase sessions.

The first correction implemented in Notebook 2 is therefore based on the logic: **SMOTE + AdaBoost + classification algorithm**. We will use SMOTE as a data-level approach to rebalance the classes, then combine it with AdaBoost as an algorithm-level ensemble method.

In our notebook, SMOTE directly targets the cause of the failure by increasing the representation of the minority class in the training data. Instead of duplicating existing purchase sessions, SMOTE generates synthetic Revenue=True examples by interpolating between neighboring minority-class samples in the transformed feature space. This gives the classifier more positive examples during learning and reduces the dominance of the majority class.

AdaBoost complements this correction from the algorithmic side. It trains a sequence of weak learners and gives more importance to examples that were misclassified in previous rounds. Since minority-class examples are more likely to be difficult under imbalanced conditions, AdaBoost can focus additional learning capacity on the decision regions where buyers are harder to detect. The correction therefore attacks the failure from two directions: **SMOTE modifies the training distribution**, while **AdaBoost focuses the classifier on difficult examples**.

The AdaBoost base estimator used in the notebook is a DecisionTreeClassifier, with shallow trees such as max_depth = 1 or max_depth = 2. Decision trees are non-linear classifiers, and no linear or logistic model is used in this correction.

4.1.2 Pipeline architecture and leakage prevention

The correction is implemented using an imblearn pipeline to avoid data leakage. The pipeline follows this structure:

Preprocessing → SMOTE → AdaBoostClassifier

The preprocessing step applies the same preparation logic as the baseline notebook: numerical variables are scaled, categorical variables are one-hot encoded, and skewed behavioral variables such as duration and PageValues are handled consistently. This preprocessing is fitted only on the training data.

SMOTE is then applied inside the pipeline, only during fit(). This is essential: SMOTE must never be applied before the train/validation/test split, and it must never be fitted on validation or test data. If SMOTE were applied before splitting, synthetic examples could leak information from the validation or test set into training, invalidating the comparison.

With this design, the validation and test sets remain untouched. They are only transformed using preprocessing fitted on the training set, and they are never resampled. The synthesized minority-class examples exist only inside the training process. This makes the before/after comparison between the untreated model and the corrected model valid, because the improvement cannot be attributed to leakage from the evaluation data.

4.1.3 Hyperparameter Search

Hyperparameter selection was performed on the **validation set only**, and the test set was kept untouched until the final evaluation. The selection criterion was **PR-AUC on the validation set**, because the task is imbalanced and PR-AUC is more informative than accuracy for evaluating the minority class Revenue=True.

The hyperparameter search was intentionally lightweight. For the standard AdaBoost benchmark using a DecisionTreeClassifier base estimator, two configurations were tested: n_estimators=100, learning_rate=1.0, max_depth=1 and n_estimators=200, learning_rate=0.5, max_depth=2. The second configuration obtained the best validation PR-AUC among the corrected models and was therefore selected for the final test-set evaluation.

For the three model families used in the baseline notebook, the correction was implemented in a model-specific way. The Random Forest wrapper was compatible with AdaBoost and was tested as SMOTE + AdaBoost + Random Forest. The MLP wrapper was not compatible with AdaBoost because of sample_weight support issues, so the notebook used a documented fallback: SMOTE + MLP. LightGBM was tested inside AdaBoost and produced a compatible corrected pipeline in the notebook, although this remains conceptually less standard because LightGBM is already a boosting model.

Thus, the validation stage compared both the untreated models and their imbalance-aware corrected counterparts. The final model was selected using validation PR-AUC, then evaluated exactly once on the test set.

4.1.4 Validation Results: Before and After Correction

Before using the test set, the models were compared on the validation set. This table is the main before/after comparison across the three model families.

Table 4 — Validation-Set Comparison: Untreated vs. Corrected Models

Model Family	Pipeline	Accuracy	Precision (True)	Recall (True)	F1 (True)	ROC-AUC	PR-AUC	Interpretation
Random Forest	Baseline, no SMOTE	0.9033	0.7799	0.5302	0.6313	0.9213	0.7385	Strong precision, but limited recall
Random Forest	SMOTE + AdaBoost + RF	0.8878	0.6616	0.5748	0.6152	0.9061	0.6734	Recall improves slightly, but PR-AUC decreases
MLP	Baseline, no SMOTE	0.8955	0.7086	0.5617	0.6266	0.9257	0.7154	Moderate recall and PR-AUC
MLP	SMOTE + MLP fallback	0.8714	0.5781	0.6509	0.6123	0.9036	0.6363	Recall improves, precision and PR-AUC decrease
LightGBM	Baseline, no SMOTE	0.9009	0.7179	0.6011	0.6543	0.9268	0.7388	Best untreated validation PR-AUC
LightGBM	SMOTE + AdaBoost + LightGBM	0.8800	0.6183	0.6037	0.6109	0.8748	0.6130	Recall stable, but PR-AUC decreases
AdaBoost + DecisionTree	SMOTE + AdaBoost + DT, n=200, lr=0.5, depth=2	0.8755	0.5821	0.7165	0.6424	0.9169	0.6872	Best corrected model by validation PR-AUC
AdaBoost + DecisionTree	SMOTE + AdaBoost + DT, n=100, lr=1.0, depth=1	0.8767	0.5844	0.7270	0.6480	0.9100	0.6561	Highest corrected recall, lower PR-AUC

4.1.5 Trade-off Discussion

The correction with **SMOTE + AdaBoost** produces a clear trade-off rather than a uniform improvement across all metrics. On the test set, the best untreated model, **LightGBM**, achieved an accuracy of **0.9033**, a precision of **0.7340**, a recall of **0.5995**, an F1-score of **0.6599**, and a PR-AUC of **0.7482**. After correction, the selected **SMOTE + AdaBoost** model reached an accuracy of **0.8878**, a precision of **0.6159**, a recall of **0.7513**, an F1-score of **0.6769**, and a PR-AUC of **0.6824**. The main gain is therefore on **Recall_true**. The corrected model detects **75.13%** of actual buyers, compared with **59.95%** for the untreated LightGBM model. This corresponds to a recall improvement of **+0.1518**. In terms of false negatives, the number of missed buyers decreases from **153 out of 382** to **95 out of 382**. This means the correction allows the model to recover **58 additional purchase sessions** that were previously missed.

However, this improvement comes with costs. Precision decreases from **0.7340** to **0.6159**, meaning that the corrected model produces more false positives: it predicts more sessions as purchases, but not all of them are actual buyers. Global accuracy also decreases from **0.9033** to **0.8878**, because the model is no longer as conservative toward the majority class Revenue=False. PR-AUC also decreases from **0.7482** to **0.6824**, which shows that the corrected model does not improve the overall ranking quality across all thresholds. Instead, it shifts the operating behavior toward higher buyer detection at the default threshold.

From a business perspective, this trade-off can still be valuable. In an e-commerce context, a false negative means that an actual buyer is missed and may not receive appropriate targeting, recommendation, or follow-up. A false positive, on the other hand, means that a non-buyer may receive unnecessary engagement. If the cost of missing a real buyer is higher than the cost of targeting a non-buyer, then the recall gain is meaningful despite the loss in precision and accuracy. The correction should therefore not be interpreted as universally better, but as **more aligned with a recall-oriented business objective**.

4.2 Second Experiment: PSO + SMOTE + AdaBoost

4.2.1 Motivation and Hypothesis

A second experiment was initially planned to extend the correction pipeline by adding a **Particle Swarm Optimization (PSO)** feature selection step before **SMOTE + AdaBoost**. The objective was to test whether selecting a smaller subset of relevant features before generating synthetic minority-class examples could improve the detection of Revenue=True.
The motivation was based on two competing hypotheses. First, PSO could help if some features were noisy, redundant, or weakly related to the minority-class signal. In that case, removing irrelevant dimensions before SMOTE could prevent the generation of synthetic examples along uninformative directions and potentially improve minority-class recall. Second, PSO could hurt performance if some features, although weak individually, carried complementary information for identifying purchase sessions. Removing such features could impoverish the minority-class representation and reduce recall.

However, this experiment was not completed in the final version of the project due to implementation and stability issues. In particular, combining PSO-based feature selection with SMOTE and AdaBoost introduced additional complexity, longer execution time, and compatibility difficulties in the notebook workflow. Therefore, we decided not to include PSO results in the final experimental comparison, in order to avoid reporting unreliable or insufficiently validated conclusions.

We keep this experiment as a planned extension rather than a validated result. If implemented in future work, it should be evaluated with the same anti-leakage protocol as the rest of the project: PSO feature selection fitted only on the training set, SMOTE applied only after feature selection and only on the training data, validation used for model selection, and the test set used exactly once for final evaluation.

5. Threats to Validity

Scientific rigor requires a self-critical analysis of the limits of our conclusions. Even though the experiments show a clear class imbalance failure and a meaningful improvement in buyer detection after correction, the results should not be interpreted as absolute proof. They depend on the dataset, the split, the chosen models, the preprocessing protocol, and the evaluation metrics. This section discusses the main reasons why our conclusions could be incomplete or partially wrong.

5.1 Random Variability Across Seeds

All experiments were made reproducible by fixing random_state = 42. This seed was used for the train/validation/test split, the model initialization when applicable, SMOTE sampling, AdaBoost, Random Forest, MLP initialization, and LightGBM randomness. This makes the reported results reproducible: running the notebooks again under the same environment should produce the same or very similar outputs.

However, using a single seed does not prove that the observed behavior is stable across different random splits or initializations. For example, the selected best untreated model was **LightGBM**, with a validation PR-AUC of **0.7388**, while the corrected model selected on validation was **SMOTE + AdaBoost + DecisionTree**, which improved test recall from **0.5995** to **0.7513**. This recall gain is substantial, but we cannot fully exclude that its magnitude depends partly on the particular split produced by seed 42.

A stronger validation protocol would repeat the full pipeline across several seeds, for example 0, 1, 7, 13, 42, and report the mean and standard deviation of each metric. This would show whether the correction consistently improves Recall_true, or whether the improvement varies strongly depending on the data split. If the standard deviation were large compared with the observed gain, the conclusion would be weaker. This repeated-seed analysis was not performed in the current project due to scope and computational constraints, so this remains an acknowledged limitation.

5.2 Test Set Size and Statistical Significance
The test set contains approximately **2,466 sessions**, corresponding to 20% of the full dataset. Because the positive class represents about **15.5**% of the data, the test set contains **382 purchase sessions**. Minority-class metrics such as Recall_true, Precision_true, and F1_true are therefore computed over these 382 positive examples, not over the full test set.

In the final comparison, the best untreated model missed **153 out of 382** buyers, while the corrected model missed **95 out of 382** buyers. This means that the correction recovered **58 additional purchase sessions**, improving recall from **0.5995** to **0.7513**. This is a practically meaningful difference: the corrected model detects a much larger fraction of actual buyers. However, we did not compute formal statistical confidence intervals. Without a paired bootstrap, McNemar's test, or confidence intervals around recall and F1-score, we cannot quantify precisely how statistically reliable this difference is. The improvement is directionally strong, but the report should not claim perfect certainty. A more rigorous evaluation would estimate uncertainty around the test metrics and verify whether the difference between the untreated and corrected models remains significant under resampling of the test set.

5.3 Confounding Effects

Although the correction improves the detection of Revenue=True, the observed changes cannot be attributed to a single factor with perfect certainty. The corrected notebook does not modify only one element of the baseline pipeline. Several components change at the same time, and each of them may influence the final metrics.

First, **SMOTE changes the training distribution** by generating synthetic minority-class examples. This directly targets the cause identified in the controlled degradation experiment: the scarcity of positive examples during training. However, SMOTE does not only increase the number of buyers in the training set; it also changes the geometry of the minority class in feature space. The synthetic examples may make the decision boundary smoother, but they may also create artificial samples that do not perfectly correspond to real web sessions.

Second, the correction notebook also tests **AdaBoost-based pipelines**, which introduce an algorithmic change compared with the untreated baselines. The original baseline models were **Random Forest**, **MLP**, and **LightGBM**, while the best corrected model selected on validation was **SMOTE + AdaBoost + DecisionTree**. Therefore, the final test improvement in recall cannot be attributed only to SMOTE. Part of the gain may come from the change in learning algorithm, especially because AdaBoost iteratively focuses on examples that are misclassified.

Third, the correction notebook also attempts to apply the paper-inspired logic to the same model families used in the baseline. It includes corrected variants such as **SMOTE + AdaBoost + Random Forest**, **SMOTE + AdaBoost + LightGBM**, and a fallback **SMOTE + MLP** when AdaBoost compatibility was problematic. This makes the comparison richer, but it also means the corrected setting is not a single isolated intervention. Some pipelines combine resampling and boosting, while others use SMOTE with the original classifier due to technical constraints.

This creates a confounding issue: when recall improves, the improvement may come from **SMOTE**, from **AdaBoost**, from the **base classifier**, or from the interaction between them. A cleaner causal attribution would require a stricter ablation study with separate comparisons such as:

Comparison	Purpose
Baseline LightGBM vs SMOTE + LightGBM	isolate the effect of SMOTE
Baseline AdaBoost vs SMOTE + AdaBoost	isolate the effect of SMOTE inside AdaBoost
SMOTE + DecisionTree vs SMOTE + AdaBoost + DecisionTree	isolate the effect of boosting
Baseline Random Forest vs SMOTE + Random Forest	test whether the same model improves after resampling

The notebook partially addresses this by comparing baseline models and corrected variants, but it does not fully isolate every component of the correction. As a result, the conclusion should be formulated carefully: the results show that **imbalance-aware correction improves buyer detection**, but they do not prove that **SMOTE alone** is responsible for the entire improvement.
Another possible confounding factor is feature informativeness. The EDA showed that variables such as PageValues, ProductRelated, BounceRates, and ExitRates are strongly related to

Revenue. If a feature like PageValues carries a large part of the predictive signal, then the model may perform well mainly because of this feature, while imbalance correction only adjusts the decision behavior around an already informative representation. In that case, class imbalance remains important, but it is not the only explanation for the model's performance.
Therefore, the correction should be interpreted as a practical improvement targeting the observed failure, not as a perfect causal proof that one single mechanism explains everything. The controlled degradation experiment supports the role of class imbalance, while the corrected pipeline demonstrates that resampling and boosting can reduce missed buyers. However, a complete causal decomposition would require additional ablation experiments.

5.4 Generalization Beyond This Dataset

The conclusions of this project are based on a single dataset: the **Online Shoppers Purchasing Intention** dataset. This dataset represents a specific e-commerce context, with its own user behavior, traffic sources, browsing patterns, seasonal effects, and purchase rate. In our data, the positive class Revenue=True represents approximately **15.5%** of the sessions. This imbalance level is important for our analysis, but it is not universal. Other e-commerce platforms may have much lower or higher conversion rates, for example 1%, 5%, or 30%, and the same correction strategy may behave differently under those conditions.
Therefore, the exact numerical results of our notebooks should not be generalized directly to all online shopping systems. For example, in our final test comparison, the corrected model improved buyer recall from **0.5995** to **0.7513**, reducing the number of missed buyers from **153** to **95** out of **382** purchase sessions. This is meaningful for this dataset, but the magnitude of this gain depends on the structure of the data, the feature distributions, the strength of variables such as PageValues, and the chosen model family.

What generalizes more reliably is the **methodological lesson**. The project shows that class imbalance can make global accuracy misleading, because a model can perform well on the majority class while missing many minority-class examples. It also shows that imbalance-aware corrections such as SMOTE and AdaBoost can shift the model toward better minority-class detection. However, the optimal correction, the best hyperparameters, and the precision–recall trade-off are dataset-specific.

In another e-commerce setting, SMOTE could be less effective if the minority class is extremely rare, if the feature space is noisier, or if synthetic examples do not correspond to realistic user sessions. Similarly, AdaBoost may help in one setting but overfit in another. For this reason, the conclusion should be phrased carefully: our results support the usefulness of imbalance-aware correction on this dataset, but they do not prove that the same correction will produce the same gains in every deployment context.

5.5 Data Leakage Risks

The notebooks were designed to reduce data leakage as much as possible. In both the untreated notebook and the correction notebook, the data is first split into **train**, **validation**, and **test** sets before learned preprocessing is fitted. This is important because transformations such as scaling and one-hot encoding must not learn information from validation or test data.

In the untreated notebook, the preprocessing is fitted on the training set only, then applied to validation and test sets. The same logic is used in the correction notebook. For the corrected pipelines, SMOTE is placed inside an imblearn pipeline, so it is applied only during training. This prevents the most serious leakage risk: applying SMOTE before the train/validation/test split. If

SMOTE had been applied to the full dataset, synthetic samples could have been generated using information from validation or test examples, making the evaluation artificially optimistic.
The final test set was also protected methodologically. It was not used for model selection. The best untreated model was selected using validation performance, and the best corrected model was also selected using validation metrics, mainly PR-AUC, recall, and F1-score. The final test evaluation was then performed once, comparing the best untreated model with the best corrected model. This respects the role of the test set as an unbiased final evaluation.

The deterministic log1p transformation applied to skewed variables such as PageValues, ProductRelated, and duration features does not create leakage because it does not learn parameters from the data or use the target. It is a fixed mathematical transformation. However, learned transformations such as StandardScaler and OneHotEncoder must remain fitted only on training data, which is why they are integrated into the preprocessing pipeline.

Some residual leakage risks remain. First, if the notebook cells are rerun manually in the wrong order, a practitioner could accidentally fit preprocessing or SMOTE on a larger dataset than intended. Second, because the test results were inspected after the final evaluation, any further model tuning based on those results would create soft leakage. We avoided this in the reported workflow, but it remains a practical risk in interactive notebooks.

Another limitation is that the dataset does not provide user identifiers. Therefore, we cannot verify whether sessions from the same user appear in both train and test sets. If repeated visits from the same individual are distributed across splits, a form of user-level leakage may exist. This is not something we can fully control with the available columns.

Finally, the planned PSO + SMOTE + AdaBoost experiment was not included in the final validated comparison due to implementation and stability issues. Therefore, no claim is made about leakage prevention inside PSO in the final results. If implemented later, PSO feature selection would need to be fitted strictly on the training set or on an inner training split only, never on validation or test labels.

5.6 Temporal Validity

The experimental protocol used in the main notebooks relies on a **stratified train/validation/test split** based on the target variable Revenue, rather than a chronological split. This choice is coherent with the main failure mode investigated in this project: **class imbalance**. By stratifying the split, the proportion of Revenue=True and Revenue=False is preserved across the training, validation, and test sets. This makes the comparison between untreated and corrected models more reliable for studying minority-class detection.

However, this design does not test temporal generalization. The dataset contains a Month feature, and the Online Shoppers dataset may include seasonal effects, promotional periods, special days, and changes in user behavior over time. With a stratified random split, sessions from the same month can appear in both training and test sets. For example, sessions from high-purchase months such as November may be present in both sets, allowing the model to benefit from patterns that would not necessarily be available in a real future deployment scenario.

Therefore, our conclusions should be interpreted carefully. The reported improvement in buyer detection is valid under a **stratified cross-sectional evaluation protocol**, where the goal is to isolate the impact of class imbalance while keeping the target distribution stable across splits. It does not prove that the corrected model would perform equally well when deployed on future months with different traffic patterns or purchase behavior.

If the objective were to study **temporal distribution shift**, the evaluation protocol would need to change. A chronological or month-based split would be more appropriate: the model would be trained on earlier months and tested on later months. This would test whether the model generalizes to future sessions, which may differ because of seasonality, marketing campaigns, holidays, or changes in browsing behavior.

Thus, the current protocol supports conclusions about **class imbalance failure and its correction**, but not about robustness under temporal drift. Temporal validity would require a separate experiment, such as training on earlier months and evaluating on unseen future months.

5.7 Validity of Synthetic SMOTE Samples

Although SMOTE is appropriate for addressing class imbalance, the synthetic examples it generates should not be treated as perfectly realistic web sessions. In our correction notebook, SMOTE is applied after preprocessing and only on the training set. This prevents data leakage, but it also means that the new minority-class examples are created in a transformed numerical feature space rather than as directly observable user sessions.

The first limitation is **realism**. SMOTE creates synthetic Revenue=True examples by interpolating between existing minority-class observations. For numerical variables such as PageValues, ProductRelated_Duration, or BounceRates, this interpolation can create intermediate values that are mathematically valid but not necessarily representative of real browsing behavior. For example, a synthetic session may lie between a highly engaged buyer and a weakly engaged buyer, without corresponding to a real user journey.

The second limitation concerns **categorical variables**. In our pipeline, variables such as Month, VisitorType, and Weekend are one-hot encoded before SMOTE is applied. After one-hot encoding, categories are represented as binary vectors. SMOTE may interpolate between these binary indicators and create fractional values such as 0.4 or 0.6 for a category column. These values are useful for the classifier in the transformed feature space, but they no longer have a direct semantic meaning as real categories.

A third risk is the **borderline effect**. SMOTE can generate synthetic samples near the decision boundary, especially if the minority examples are heterogeneous. This may help the model detect more buyers, but it may also increase false positives. This trade-off appears in our final test results: the corrected model improves recall from **0.5995** to **0.7513**, but precision decreases from **0.7340** to **0.6159**. Therefore, SMOTE improves minority-class detection, but it also makes the classifier less conservative when predicting Revenue=True.

For this reason, the SMOTE correction should be interpreted as a practical method for reducing missed buyers, not as a perfect reconstruction of the true minority-class distribution. It improves the training signal for Revenue=True, but the synthetic examples may not fully correspond to realistic e-commerce sessions.

5.8 Planned PSO Extension and Why It Was Not Included

A second extension was initially considered: adding a **Particle Swarm Optimization (PSO)** feature-selection step before SMOTE and AdaBoost. The idea was to test whether selecting a smaller subset of informative features could improve the quality of the synthetic minority examples and reduce noise before applying the imbalance correction.

The motivation was based on two competing possibilities. On one hand, PSO could help if some features are noisy or redundant. In that case, removing them before SMOTE could prevent the algorithm from generating synthetic buyers along irrelevant feature dimensions. On the other hand, PSO could hurt performance if it removes weak but useful features that contribute to identifying Revenue=True. This is especially important in imbalanced classification, where minority-class signals may be distributed across several features rather than concentrated in one obvious variable.
However, this PSO experiment was **not included in the final validated comparison**. The reason is that combining PSO, SMOTE, and AdaBoost introduced additional implementation complexity and stability issues. PSO is stochastic, computationally heavier, and sensitive to the choice of swarm size, number of iterations, and fitness function. Because the project already contained a complete controlled degradation experiment and a validated SMOTE + AdaBoost correction, we chose not to report PSO results that were not sufficiently stable or fully validated.

If this extension were implemented in future work, it would require a strict anti-leakage design. PSO feature selection would need to be fitted only on the training data, ideally using an inner split of X_train. Validation and test labels should never be used to select features. The full PSO experiment should also be repeated across several random seeds to check whether the selected feature subset is stable. If PSO selects very different features across runs, then conclusions about feature relevance would need to be treated with caution.

Thus, PSO remains a possible future improvement, but not a validated result of the current project. The final conclusions are therefore based only on the experiments that were actually executed and evaluated: the untreated models, the controlled degradation experiment, and the SMOTE + AdaBoost correction.

6. Conclusion and What We Learned

6.1 Summary of Findings

This project investigated **class imbalance failure** on the Online Shoppers Purchasing Intention dataset using a structured scientific approach: first making the failure visible, then formulating a causal hypothesis, testing it through a controlled experiment, and finally evaluating a correction. The objective was not simply to maximize predictive performance, but to understand why the model fails on the minority class Revenue=True and whether the correction actually targets the cause.

6.1.1 The failure was made visible through the right metrics

The untreated non-linear models — **Random Forest**, **MLP**, and **LightGBM** — achieved high global accuracy, around **89.6% to 90.3%**, but their recall on Revenue=True remained much lower. For example, the best untreated model, **LightGBM**, reached a validation accuracy of **0.9009** and a validation PR-AUC of **0.7388**, but its recall on the positive class was only **0.6011**. On the test set, the same failure appeared clearly: the best untreated LightGBM missed **153 out of 382** actual buyers, meaning that about **40.1%** of purchase sessions were classified as non-purchase.
This shows that the model did not fail randomly. It failed asymmetrically: it was much stronger on the majority class Revenue=False than on the minority class Revenue=True. The key contribution

of the first notebook was therefore to make the failure measurable using **Recall_true**, **F1_true**, **PR-AUC**, and the confusion matrix, instead of relying on accuracy alone.

6.1.2 The cause was tested, not only assumed

The low recall was not attributed to class imbalance only because the dataset contains approximately **15.5%** positive examples. A controlled degradation experiment was added in Notebook 2 to test the causal hypothesis more directly. In this experiment, the validation set, test set, preprocessing pipeline, model family, negative examples, and random seed were kept fixed. Only the number of positive training examples was progressively reduced.
The results showed a clear degradation pattern. When **100%** of positive examples were retained, the training positive rate was **15.64%** and Recall_true was **0.575**. When only **10%** of positive examples were retained, the training positive rate dropped to **1.81%** and recall collapsed to **0.066**. The same trend appeared for F1_true, which decreased from **0.613** to **0.121**. This dose-response behavior supports the hypothesis that the scarcity of positive examples is causally involved in the minority-class detection failure.

6.1.3 The correction targeted the identified cause

The correction tested in Notebook 2 was based on **SMOTE + AdaBoost**, inspired by the paper of Kurniawan et al. The goal was to address the underrepresentation of Revenue=True during training. SMOTE increases the effective number of minority-class examples by generating synthetic positive samples, while AdaBoost focuses learning on examples that are difficult to classify.

The final test comparison showed that the correction improved buyer detection. The untreated LightGBM model reached a recall of **0.5995** and missed **153** buyers. The corrected **SMOTE + AdaBoost** model reached a recall of **0.7513** and missed only **95** buyers. This means the correction recovered **58 additional purchase sessions**. However, the improvement came with a trade-off: precision decreased from **0.7340** to **0.6159**, accuracy decreased from **0.9033** to **0.8878**, and PR-AUC decreased from **0.7482** to **0.6824**. Therefore, the correction should be interpreted as a recall-oriented improvement, not as a universal improvement on every metric.

6.2 What This Project Taught

The main lesson of this project is not that one algorithm is always better than another. The deeper lesson is that machine learning evaluation must be diagnostic. A model can appear successful according to global accuracy while still failing on the class that matters most for the application. In this project, the untreated models reached around **90% accuracy**, but the best untreated model still missed **153 out of 382** buyers on the test set. This shows that accuracy alone can hide the real failure: poor detection of Revenue=True.

Several more specific lessons emerged from the experimental design:

First, **metrics must match the cost of the problem**. In this dataset, a false negative means that an actual buyer is classified as a non-buyer. From a business perspective, this is costly because the system fails to identify a session that could benefit from targeted engagement or follow-up. For this reason, Recall_true, F1_true, and PR_AUC were more informative than accuracy. The corrected model improved recall from **0.5995** to **0.7513**, reducing missed buyers from **153** to **95**. Even though accuracy and precision decreased, the correction became more aligned with a recall-oriented objective.

Second, **class imbalance must be tested, not only observed**. It would have been weak to simply say that the dataset has around **15.5%** positive examples and therefore imbalance is the cause of low recall. The controlled degradation experiment made the causal argument stronger. When the positive training rate was reduced from **15.64%** to **1.81%**, Recall_true collapsed from **0.575** to **0.066**, and F1_true dropped from **0.613** to **0.121**. This dose-response pattern showed that the availability of positive examples directly affects the model's ability to detect buyers.

Third, **leakage prevention is a structural requirement, not just a good intention**. The notebooks were designed around an anti-leakage protocol: the data was split before learned preprocessing, scalers and encoders were fitted only on training data, SMOTE was placed inside an imblearn pipeline, and the test set was reserved for final evaluation. These choices were essential because applying SMOTE before splitting, or fitting preprocessing on the full dataset, would have made the validation and test results artificially optimistic.

Fourth, **a correction must be connected to the cause it claims to address**. SMOTE + AdaBoost was not used simply because it is a common recipe for imbalanced datasets. It was justified because the controlled experiment showed that scarcity of positive examples harms minority-class recall. SMOTE directly increases minority-class representation during training, while AdaBoost gives more attention to difficult examples. The correction therefore targets the training mechanism that produced the observed failure.

Fifth, **stronger models are not automatically robust to structural data problems**. Random Forest, MLP, and LightGBM are all non-linear models, yet all of them showed the same general failure pattern when trained without imbalance correction: high global performance but weaker detection of Revenue=True. This demonstrates that model complexity does not remove the need to understand the data distribution. Even powerful models can inherit and amplify imbalance-driven bias.

Finally, **acknowledging limitations improves the credibility of the analysis**. The results are meaningful, but they are not absolute. They depend on one dataset, one split strategy, one random seed, and a specific correction design. The correction improved recall, but also reduced precision and PR-AUC. Therefore, the conclusion is not that SMOTE + AdaBoost is universally better, but that it shifts the model toward better buyer detection under the assumptions of this experiment. A rigorous ML analysis is not about claiming a perfect result; it is about showing what was tested, what was learned, and what remains uncertain.

6.3 Open Questions

This project answers the main research question under a controlled stratified evaluation protocol, but it also opens several directions for future work.

A first open question concerns **temporal generalization**. The current train/validation/test split is stratified by Revenue, which is appropriate for studying class imbalance. However, the dataset contains a Month variable, suggesting possible seasonal effects. A chronological or month-based split could reveal whether the model also suffers from temporal distribution shift. In that setting, the model would be trained on earlier months and evaluated on future months, which may contain different user behavior, marketing campaigns, or purchase patterns.

A second question concerns the choice of correction method. This project used **SMOTE + AdaBoost** to increase minority-class detection, and the correction improved recall from **0.5995** to **0.7513** on the test set. However, this improvement came with lower precision and lower PR-AUC.

It would therefore be useful to compare this correction with other imbalance-aware methods, such as class weighting, cost-sensitive learning, threshold tuning, or focal-style losses. These alternatives might improve recall while making fewer assumptions about the geometry of the minority class.

A third question concerns the role of highly predictive features such as PageValues. The EDA suggests that PageValues is strongly associated with Revenue=True. This raises an important question: is PageValues a stable behavioral signal, or could it behave like a shortcut that works well in this dataset but generalizes poorly to other time periods, websites, or user segments? Testing this would require a second failure-mode investigation based on feature removal, permutation, or subgroup evaluation.

A fourth question concerns statistical robustness. The final test set contains **382 positive examples**, and the correction reduced false negatives from **153** to **95**. This is practically meaningful, but the project did not compute confidence intervals or run repeated-seed evaluations. A stronger validation would repeat the full pipeline across multiple random seeds and use bootstrap confidence intervals or McNemar’s test to assess whether the recall improvement is statistically reliable.

Finally, the planned **PSO + SMOTE + AdaBoost** extension remains an open direction. Feature selection before SMOTE could potentially reduce noise and improve the quality of synthetic minority examples. However, it could also remove weak but useful minority-class signals. Because this experiment was not stable enough to include in the final validated comparison, it should be treated as future work rather than as a conclusion of this project.
These open questions reflect the scientific posture required by the project: the goal is not only to produce a better model, but to understand what the result proves, what it does not prove, and what experiments would be needed to make the conclusion stronger.